11.18 Classwork: Series — Markscheme

1. [Maximum mark: 5]

(a) Find S_n. [1]

$$S_n = \frac{10^n - 1}{9}$$

A1

(b) Show that $S_1 + S_2 + \cdots + S_n = \dfrac{10(10^n - 1) - 9n}{81}$. [4]

$$S_1 + S_2 + \cdots + S_n = \frac{10 - 1}{9} + \frac{10^2 - 1}{9} + \cdots + \frac{10^n - 1}{9}$$

A1

$$= \frac{(10 + 10^2 + \cdots + 10^n) - n}{9}$$

M1

$$10 + 10^2 + \cdots + 10^n = \frac{10(10^n - 1)}{10 - 1}, \qquad (-1) + (-1) + \cdots + (-1) = -n$$

A1

$$S_1 + S_2 + \cdots + S_n = \frac{\frac{10(10^n-1)}{9} - n}{9} = \frac{10(10^n - 1) - 9n}{81}$$

A1 AG

2. [Maximum mark: 8]

(a)(i) Show that the area of F_n is $20\left(\frac{9}{4}\right)^{n-1}$ cm^2. [2]

The width and height each have common ratio $\frac{3}{2}$, so the areas form a geometric sequence with first term 20 and common ratio

$$\left(\frac{3}{2}\right)^2 = \frac{9}{4}.$$

M1 A1

$$A_n = 20\left(\frac{9}{4}\right)^{n-1}$$

AG

(a)(ii) Find the mean area. [3]

$$\sum_{n=1}^{10} A_n = \frac{20\left(\left(\frac{9}{4}\right)^{10} - 1\right)}{\frac{9}{4} - 1}$$

M1

$$= 16\left(\left(\frac{9}{4}\right)^{10} - 1\right)$$

A1

Therefore the mean area is

$$\frac{1}{10} \cdot 16\left(\left(\frac{9}{4}\right)^{10} - 1\right) = \frac{8}{5}\left(\left(\frac{9}{4}\right)^{10} - 1\right) \text{ cm}^2.$$

A1

(b) Find the median area. [3]

The median is the mean of the 5th and 6th areas. **M1**

$$\text{median} = \frac{20\left(\frac{9}{4}\right)^4 + 20\left(\frac{9}{4}\right)^5}{2}$$

A1

$$= \frac{20\left(\frac{9}{4}\right)^4\left(1 + \frac{9}{4}\right)}{2} = \frac{65}{2}\left(\frac{9}{4}\right)^4 \text{ cm}^2.$$

A1

3. [Maximum mark: 16]

(a) Write down t_3. [1]

$$t_3 = 15$$

A1

(b) Find t_4. [2]

$$t_4 = 15 + 11$$

M1

$$t_4 = 26$$

A1

(c) Show that $t_n = \dfrac{n(3n+1)}{2}$. [3]

The row totals form the arithmetic sequence

$$2 + 5 + 8 + 11 + \cdots .$$

M1

$$t_n = \frac{n}{2}\left(2(2) + 3(n-1)\right)$$

M1

$$t_n = \frac{n}{2}(3n+1) = \frac{n(3n+1)}{2}$$

A1 AG

(d) Find the maximum number of rows from 14 full packs. [3]

$$\frac{n(3n+1)}{2} \leq 14(52) = 728$$

M1

Solving gives $n = 21.8642\ldots$, or checking:

$$t_{21} = 672, \qquad t_{22} = 737.$$

A1

The maximum number of rows is 21. **A1**

(e) Find the minimum number of rows using full packs with no cards left over. [2]

We need

$$\frac{\frac{1}{2}n(3n+1)}{52} \in \mathbb{Z}.$$

M1

The minimum number of rows is 13. **A1**

(f) Find the minimum number of cards for height more than 2 metres. [5]

The vertical height of one triangular row is

$$88\sin 60^\circ = 44\sqrt{3}\text{ mm}.$$

M1 A1

We need

$$44n\sqrt{3} > 2000.$$

M1

$$n > 26.2431\ldots,$$

so the minimum number of rows is 27. **A1**

$$t_{27} = \frac{27(3(27)+1)}{2} = 1107.$$

A1

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4. [Maximum mark: 12]

(a)(i) Monthly interest for Andy. [1]

$$5000(0.00315) = 15.75$$

A1

(a)(ii) Annual interest for Andy. [1]

$$12(15.75) = 189$$

A1

(a)(iii) Andy's amount after n years. [1]

$$5000 + 189n$$

A1

(a)(iv) Show Andy has \$5945 after 5 years. [1]

$$5000 + 189(5) = 5945$$

A1 AG

(b)(i) Jess's account after 5 years. [2]

$$5000(1.03)^5$$

M1

$$= 5796.37\ldots \approx \$5796$$

A1

(b)(ii) Jess's interest over 5 years. [1]

$$5796.37\ldots - 5000 = 796.37\ldots \approx \$796$$

A1

(c)(i) Jess's amount after n years. [2]

Use the compound-interest model with annual compounding. **M1**

$$5000(1.03)^n$$

A1

(c)(ii) Smallest number of complete years. [3]

$$5000(1.03)^n > 5000 + 189n$$

M1

Solving or checking values:

$$5000(1.03)^{16} = 8023.53\ldots < 8024, \qquad 5000(1.03)^{17} = 8264.23\ldots > 8213.$$

M1

Therefore the smallest number of complete years is 17. **A1**